ClinGen's Variant Curation Interface (VCI)

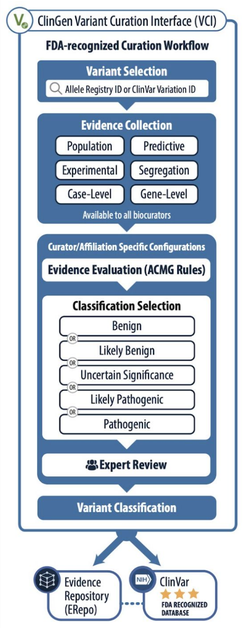
Mandell, M.E.[1], Preston, C.G.[1], Wright, M.W.[1], Dalton, K.P.[1], Cheung, G.[1], Wulf, B.[1], DiStefano, M.[3,4], Harrison, S.M.[3], Strande, N.T.[4], Ross, J.E.[6], Dziadzio, H.[3], Klein, C.J.[1], Shapira, R.[1], Keseler, I.M.[1], Ritter, D.I.[5], Shah, N.[2], Riehle, K.[2], Milosavljevic, A.[2], Plon, S.E.[2,5], Klein, T.E.[1], on behalf of the Clinical Genome Resource (ClinGen)

[1]Department of Biomedical Data Science, Stanford University School of Medicine, Stanford, CA
[2] Molecular and Human Genetics Department, Baylor College of Medicine, Houston, TX
[3] Medical and Population Genetics, Broad Institute of MIT and Harvard, Cambridge, MA
[4] Precision Health Program, Geisinger, Danville, PA
[5] Pediatrics-Hematology-Oncology, Baylor College of Medicine, Houston, TX
[6] Department of Genetics, University of North Carolina, Chapel Hill, NC

Abstract

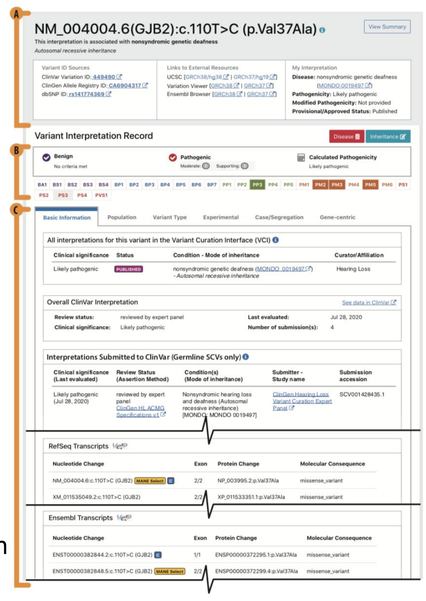
The NIH-funded Clinical Genome Resource Consortium Variant Curation Interface (ClinGen VCI) is a global, open-source variant classification platform for supporting the application of evidence criteria and classification of variants based on the ACMG/AMP sequence variant classification guidelines. To facilitate evidence-based improvements in human variant classification, the VCI is publicly available to the worldwide genomics community. The VCI is among a suite of tools developed by ClinGen, and supports an FDA-recognized human variant curation process of ClinGen Variant Curation Expert Panels (VCEPs). The variant curation workflow is intended to support dissemination of variant curations into two public repositories: the ClinGen Evidence Repository (for approved ClinGen VCEPs) and ClinVar.
ClinGen is expanding to involve more curators and teams of curators working together (affiliations) as part of an increasing scale of activities that will increase genetic variant curations across a greater number of genes. The VCI platform is built on a serverless architecture and is designed to support future enhancements and enable scalable curation workflows. Features that support curation accuracy, scaling, and transparency include an audit trail, a variant prioritization feature, and integration with ClinGen's Criteria Specification Registry (CSpec). These developments support the ClinGen mission of curating the continually-expanding space of the clinical human genome.

ClinGen's Variant Curation Process and VCI

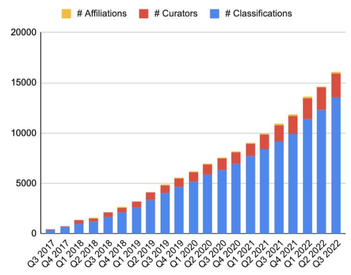


Left: Overview of the ClinGen Variant Curation Process using the VCI. Biocurators select a variant, and evaluate evidence. VCI viewers may view all evidence available for any variant using the VCI. ClinGen expert panels disseminate their final classifications through two community resources: ClinVar and the ClinGen Evidence Repository (ERepo).

Right: The VCI "Basic Information" view. The VCI has six tab views that collate and display variant information from external and internal sources to biocurators. **(A)** The top title view is always viewable and shows the variant title, and key curation information for the record. **(B)** The criteria bar displays evaluated criteria and the calculated pathogenicity. **(C)** The basic information tab displays data from ClinVar, RefSeq, and Ensembl.



VCI Growth

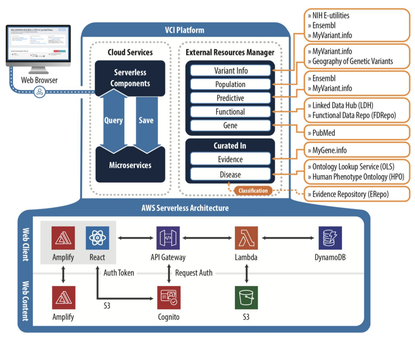


Growth in the VCI over the last 5 years

Chart of the growth of:

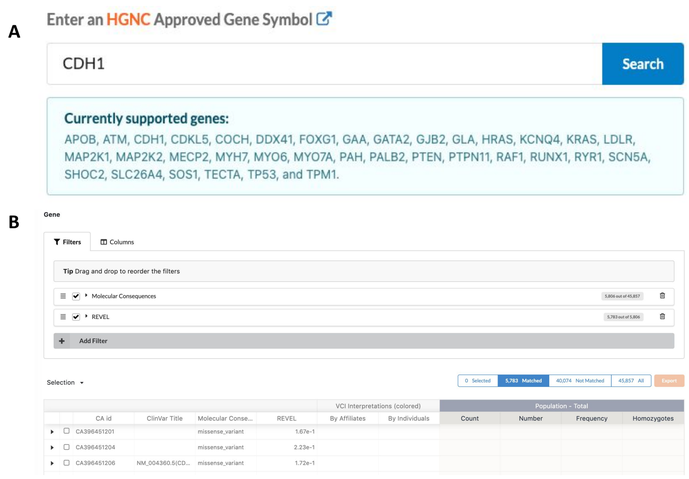
- VCI classifications (13561)
- Biocurators (2378)
- Affiliations (132)

The VCI Infrastructure



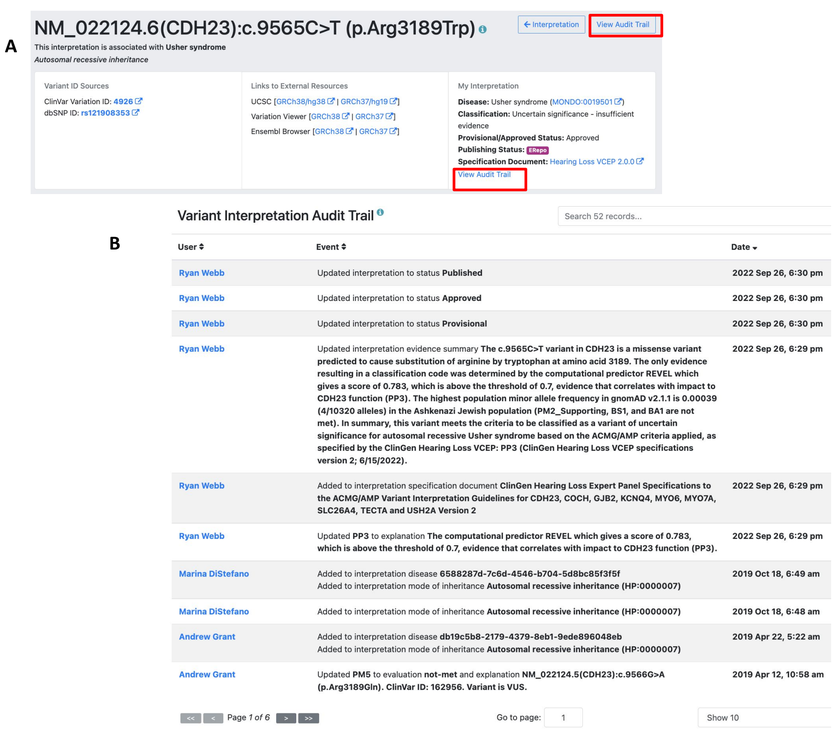
VCI Platform Components Overview Including Schema and Serverless Architecture. The platform is web-browser based and uses AWS cloud services. An External Resources Manager retrieves population, predictive, functional, and other variant and gene-level data from external sources via APIs. In addition, the user can add in curated evidence, evaluate against ACMG/AMP guidelines and save the classification for review and approval. The approved classifications are then submitted to the ClinGen Evidence Repository. All data are saved in a database via microservices and can be accessed via queries. The Amazon Cloud Services provide the microservices to store and retrieve data, while the client-side web interface is created using the React JavaScript library. This serverless architecture allows for automated scaling, as the cloud services concurrently process and load data, a key feature needed for bulk curation.

Variant Prioritization



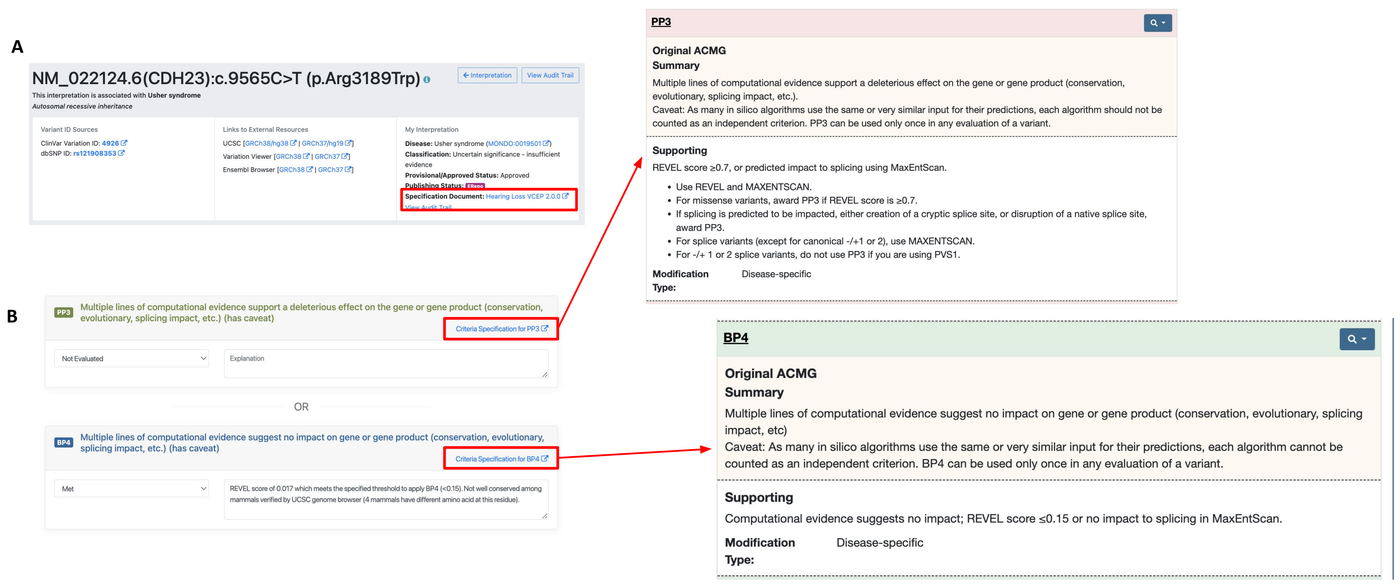
Variant Prioritization is one of several planned VCI features to enable and support bulk curation and variant assessment. **(A)** The Variant Prioritization feature allows a user to identify, search and sort on all variants and associated data for a curated list of genes (>100 genes will be supported in early 2023). The data is provided by the ClinGen Linked Data Hub. **(B)** Curators filter their variants on data such as Molecular Consequences, ClinVar status, VCI status, gnomAD data and the metapredictor REVEL.

The Audit Trail



The Audit Trail. (A) The audit trail can be accessed from two places in the variant header (red boxes) in the VCI. **(B)** The audit trail displays a searchable, click-by-click, and time stamped record of all curation actions for that variant.

VCI and Criteria Specification Registry Integration



VCI and Criteria Specification Registry Integration.

The VCI has been integrated with ClinGen's Criteria Specification Registry (CSpec) https://cspec.genome.network/cspec/ui/svi/, for approved ClinGen Variant Curation Expert Panels (VCEPs).

(A) VCEPs can add a specification document to their curation record.

(B) Users will then will see links as they are doing their curation work that will lead them to the specific section of the CSpec that has the ClinGen Sequence Variant Interpretation (SVI) approved specification for their curation work.

The CSpec details are sent along to ClinVar when the curation is submitted, and eventually we plan to display the CSpec language directly in the VCI. This feature supports consistency and transparency in curation.

Acknowledgement: Authors would like to thank all the members of ClinGen Variant Curation Expert Panels, Sequence Variant Interpretation VCEP Review Committee, Clinical Domain Working Group Oversight Committee and ClinGen Data Exchange. **Sources of support**: This research was funded by grants from the National Human Genome Research Institute (NHGRI) of the National Institutes of Health (U24HG009649, U41HG009649).

Questions? Comments? Ideas? Contact us at *vci@clinicalgenome.org*